

LEARNOVA MOBILE APPLICATION

Complete Server Setup, IP Discovery & APK Connection Guide

1. System Architecture Overview

Learnova connects your mobile device directly to the Django REST backend server running on your computer. Because both devices communicate over your local Wi-Fi router, the mobile app requires your computer's local **IPv4 address** (e.g., 10.143.206.252) instead of 127.0.0.1 or localhost.

Mobile Phone (Learnova APK) Running on Android Phone Target: http://<PC-IP>:8000/api	↔ Wi-Fi Router ↔ Local WLAN Network	Development PC (Django Server) Running on 0.0.0.0:8000 Wi-Fi IPv4: 10.143.206.252
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2. How to View Your PC's Server IP Address (`ipconfig`)

Wi-Fi routers automatically assign dynamic IP addresses via DHCP. Whenever your computer disconnects or reconnects to Wi-Fi, your IP address may change. Follow these quick steps to find your current IP:

Step 1: Press **Win + R**, type **cmd** or **powershell**, and press **Enter**.

Step 2: In the black terminal window, type the following command and press Enter:

```
ipconfig
```

Step 3: Scroll through the output and look for the section titled **Wireless LAN adapter Wi-Fi**:

```
Wireless LAN adapter Wi-Fi:
  Connection-specific DNS Suffix  . : 
  Link-local IPv6 Address . . . . . : fe80::7e3b:baed:bd80:9d26%7
  IPv4 Address. . . . . : 10.143.206.252    <--- THIS IS YOUR SERVER IP
  Subnet Mask . . . . . : 255.255.255.0
  Default Gateway . . . . . : 10.143.206.18
```

■■ **Note:** Always copy the **IPv4 Address** from **Wireless LAN adapter Wi-Fi**. Do NOT use 'Ethernet' or virtual adapter IPs (e.g. 192.168.56.1 from VirtualBox) when your phone is connected over Wi-Fi.

3. How to Run the Django Backend Server

To allow external devices like your physical phone to communicate with your backend, you must bind the server to **0.0.0.0:8000**. If you only run `python manage.py runserver` without specifying 0.0.0.0, Django only listens to 127.0.0.1 (localhost) and the phone cannot connect.

Step 1: Open terminal/PowerShell in the project backend folder:

```
cd "c:\Users\PERSONAL\OneDrive\Desktop\mad assignment\backend"
```

Step 2 (Optional - First Time Only): Apply database migrations and load demo seeds:

```
python manage.py migrate
python seed_data.py
```

Step 3: Start the server bound to all network interfaces on port 8000:

```
python manage.py runserver 0.0.0.0:8000
```

■■ **Note:** You should see output: Starting development server at http://0.0.0.0:8000/. Keep this terminal window OPEN while using the mobile app.

4. How to Connect the Learnova APK with the Server

Your new Learnova APK features an **Automatic Server Setup Prompt** right on the splash screen loading phase, meaning you can seamlessly connect to any new Wi-Fi IP address in seconds without reinstalling the app!

Step 1: Check Wi-Fi Connection

Ensure both your Android smartphone and your PC are connected to the **same Wi-Fi network**.

Step 2: Install and Open the APK

Install **app-release.apk** on your phone and launch **Learnova**.

Step 3: Enter IP on the Splash Screen

During app startup, a bottom sheet titled '**Connect to Server**' will appear automatically:

- **Server IPv4 Address:** Type the IPv4 address from your ipconfig (e.g., 10.143.206.252).
- **Port:** Leave as 8000.
- Tap '**Connect & Continue**'.

The app saves this IP in its secure storage, so you don't need to retype it until your Wi-Fi changes.

Step 4: Changing IP Later from the Login Screen

If your IP changes tomorrow, look at the bottom of the Login Screen. You will see a button labeled '**Server: 10.143.206.252:8000**'. Tap it at any time to update your IP without restarting or reinstalling!

5. Default Demo Login Credentials

You can use any of the pre-configured accounts to test all student, faculty, and administrative flows:

Role	User ID / Username	Password	Features Available
Student	23CSE001	student123	Assignments, Timetable, Notes, OCR Submissions

Faculty / Teacher	FAC001	faculty123	Create Tasks, Plagiarism Check, Grading & Reviews
Administrator	admin	admin123	Manage Users, System Analytics, Global Notifications

6. Quick Troubleshooting & Firewall Fixes

Issue: 'Cannot connect to server at http://...'	1. Verify both phone and PC are on the same Wi-Fi. 2. Confirm the server was started with 0.0.0.0:8000. 3. Test connection by opening Chrome on your phone and visiting http://<PC-IP>:8000/api/.
Issue: Windows Firewall blocks incoming phone requests	In Windows Search, open ' Windows Defender Firewall with Advanced Security ' → Inbound Rules → New Rule → Port → TCP 8000 → Allow the connection . This allows phones on your home/office Wi-Fi to reach port 8000.
Issue: Android shows 'Unsafe installation blocked'	Because this is a private development APK built for your project, Android Play Protect may show a warning. Tap ' More Details ' and select ' Install Anyway '.